

Designing Thinking Machines

The Future of AI with Graphs-Based AI Agents

Open Source AI Workshops

Berner Fachhochschule, Bern, May 19th

Who we are



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iCoSys + HumanTech

- Part of **HES-SO**, located in Fribourg.
- **Interdisciplinary** centers of **excellence** and national reference institutes in applied AI and complex systems.
- Develops **impactful** joint projects with **companies** and the **public** sector.

Alpine Event Contract Assistant

From raw business documents to evidence-based decisions



Goal : Build an assistant that helps event organizers assess requests and operational decisions against contractual requirements, with clear supporting evidence.

Document pack



Alpine Event Contract



Annex: scope sheet



Email exchange



Fondue Recipe



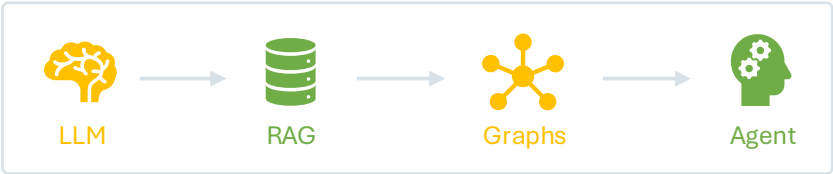
User request

“Is the proposed venue compliant?”



Answer with Evidence:

- 1) **Decision** (Suitable / Conditional / Not Suitable)
- 2) **Evidence** (Policy clauses, venue details, ...)
- 3) **Source** (Document snippets, emails)



Workshop Timeline

We build the event contract assistant by adding one capability at a time.

09:00 – 10:15	Part 01: Foundations	• LLM basics, first PydanticAI app
10:15 – 10:30	Break	
10:30 – 12:00	Part 02: RAG	• Chunking, retrieval, generation
12:00 – 13:00	Lunch	
13:00 – 15:00	Part 03: Graph-RAG	• Schema definition, graph construction
15:00 – 15:30	Break	
15:30 – 17:00	Part 04: Tools, Agents	• Tool calling, agents with multiple tools

Workshop Repository



Github: https://github.com/ovaccarelli/workshop_graph_rag

Graph_rag/

├ data/

├ exercises/

│ ├── part_01/

│ ├── part_02/

│ ├── part_03/

│ └ part_04/

├ presentation/

← this presentation

├ solutions/

← all the solutions to the exercises

├ src/

← utils

└ README.md



What is an LLM?

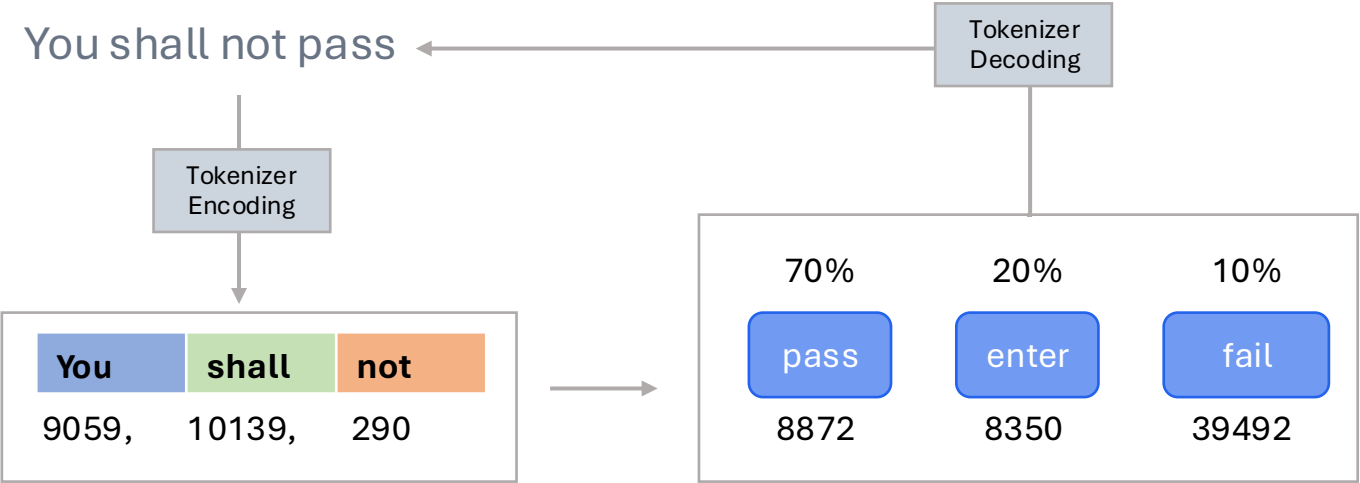
AI system trained on vast amounts of text that can understand context and generate human-like language.



Core Idea

- 1 Context in
- 2 Tokenize
- 3 Predict (next-token probabilities)
- 4 Decode & repeat

You shall not pass



Good at



Summarization/
Translation



Question
Answering



Text
Generation



Reasoning
over Context

Limits



Hallucinations



Context
Limit



No native access
to external data



Need tools
to act

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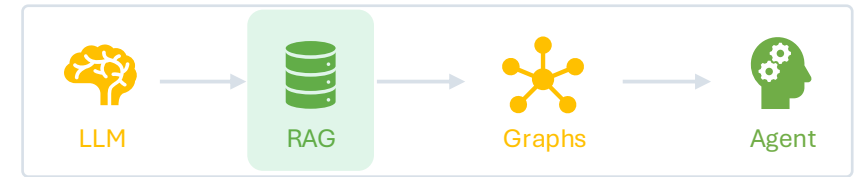
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BREAK

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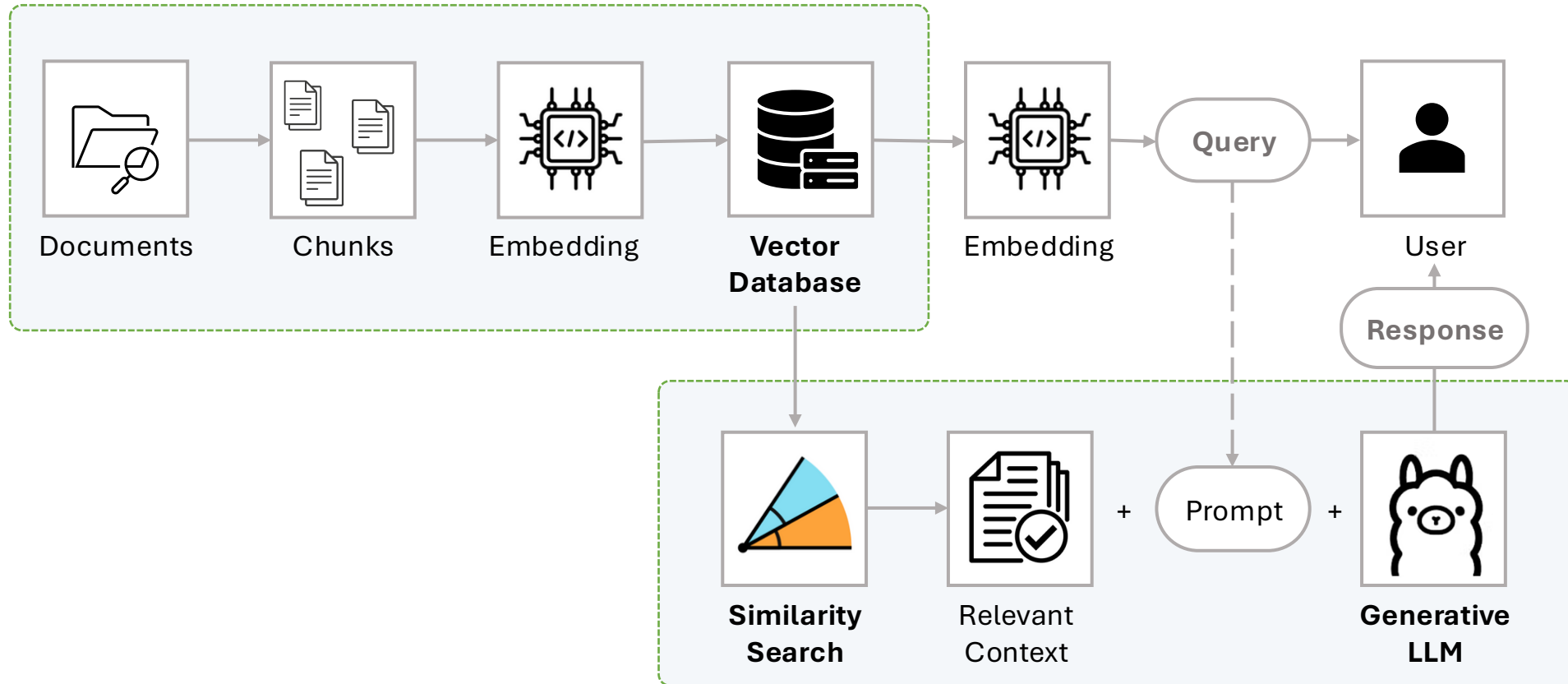
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RAG: Retrieval Augmented Generation

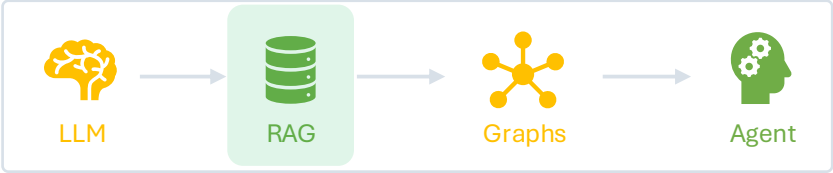
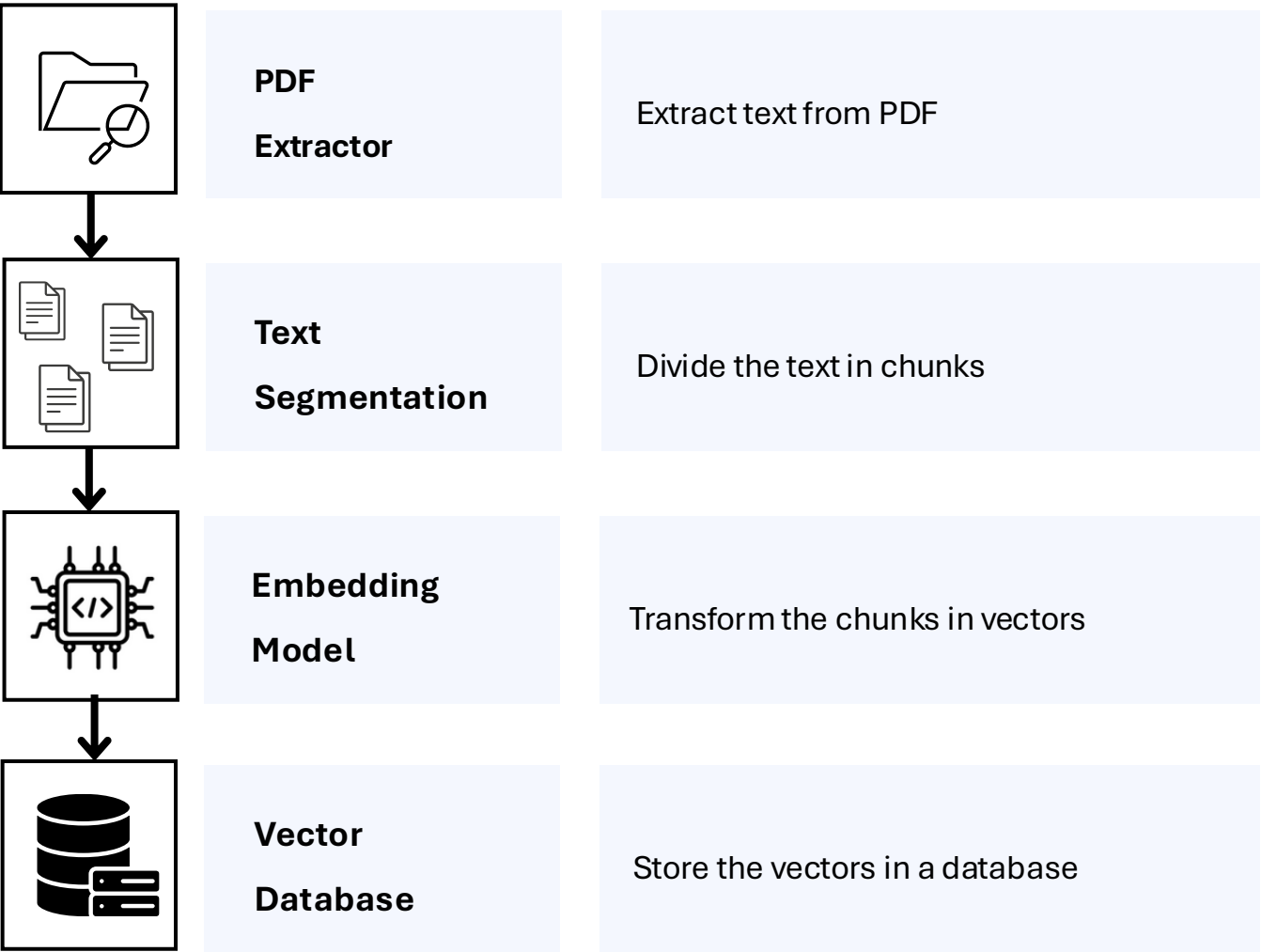
The model answers using retrieved document chunks instead of guessing.

Offline Data Preprocessing



Online Semantic Retrieval

RAG: Offline Data Preprocessing



Service Agreement

SERVICE AGREEMENT - ALPINE EVENING

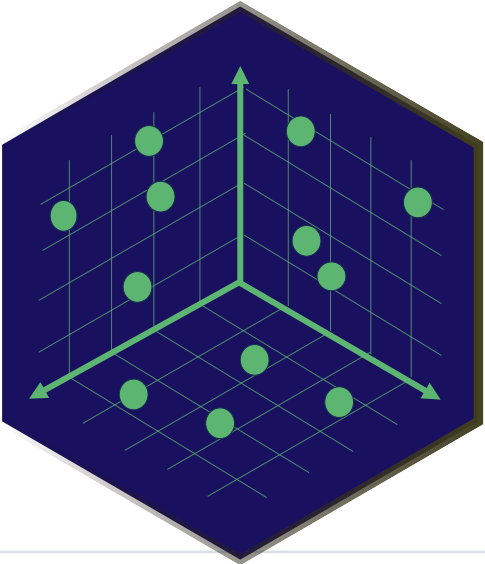
Organizer: Elven Ledger Council of Rivendell SA | Provider: Shire Hospitality Cooperative GmbH

Document ID: ELC-SHC-2026-02
Signature date: 02 May 2026
Event date: 24 May 2026

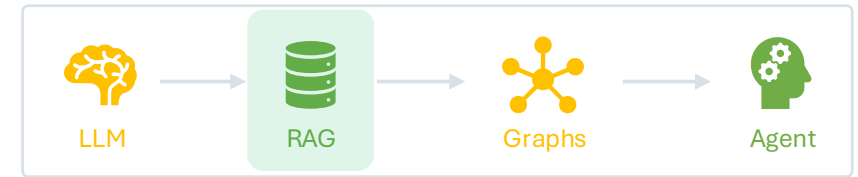
Parties
Organizer: Elven Ledger Council of Rivendell SA ("Organizer") Provider: Shire Hospitality Cooperative GmbH ("Provider")

1. Purpose
Provider shall deliver the "Alpine Evening" services (food and performance) for Organizer, as described in Annex A (Scope Sheet).

Chunk 1



RAG: Online Semantic Retrieval

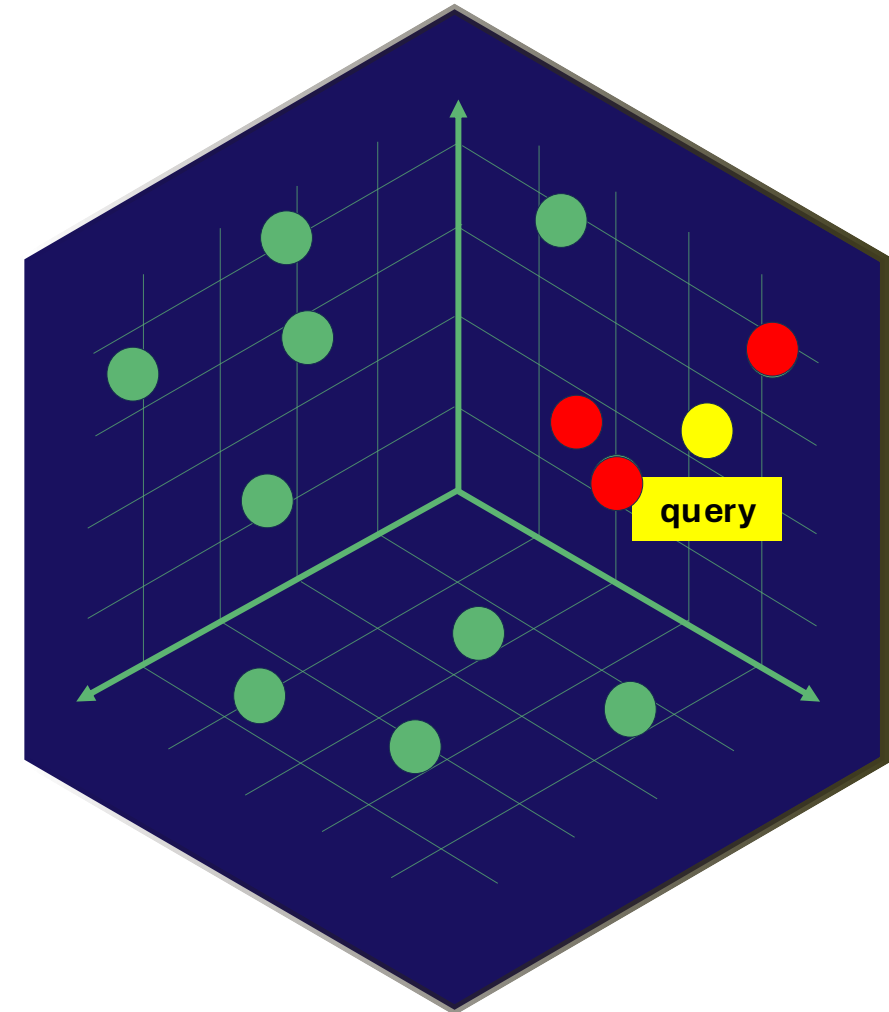
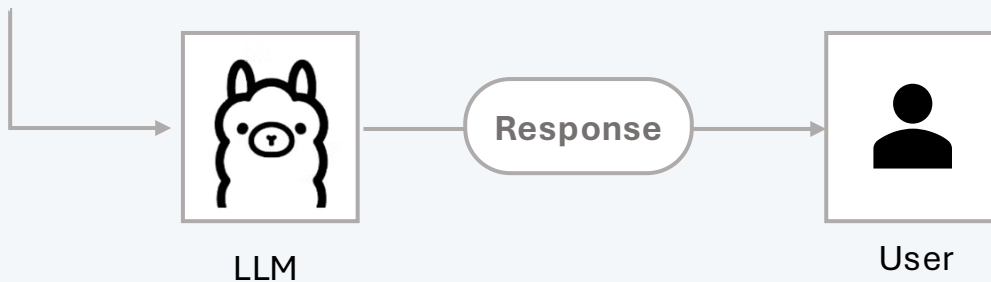


Query : also transform into a vector.

We search the K=3 chunks "similar" to the query.

- **Metric:** Cosine similarity
$$\text{Cos_sim}(\mathbf{q}, \mathbf{v}_i) = \frac{\mathbf{q} \cdot \mathbf{v}_i}{\|\mathbf{q}\| \|\mathbf{v}_i\|}$$

System Prompt + User prompt + Relevant Context



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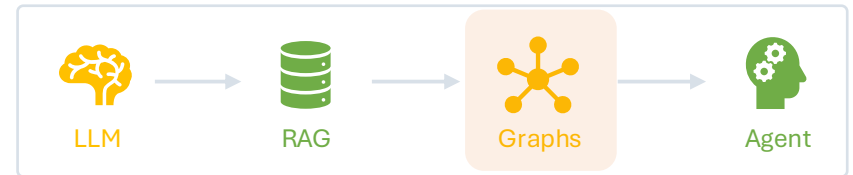
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LUNCH

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Euler's Bridges of Königsberg

From a city map to a graph: the origin story

Graph model

Nodes = places

Edges = connections

Classic question

Can we traverse every bridge exactly once? (Eulerian Trail)

Graph intuition

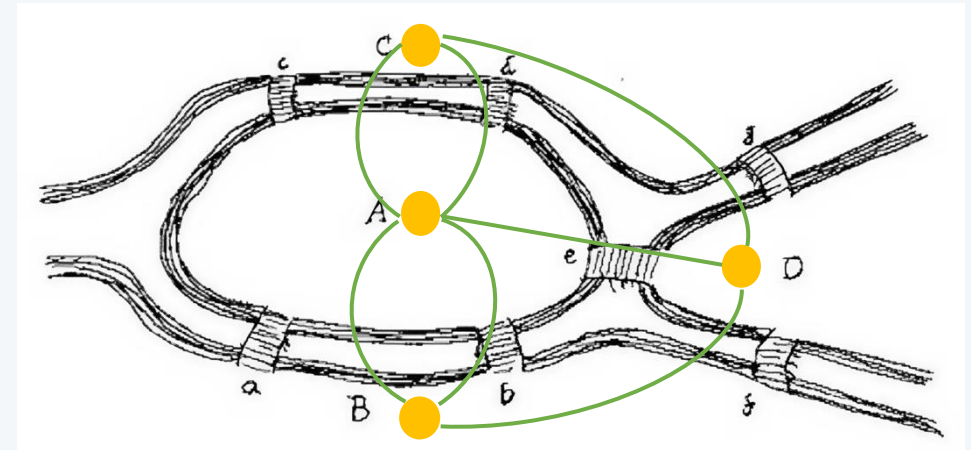
Connectivity balance: structure drives what is possible

Continuous traversal, dead ends depending on number of connections

Diagram

Abstraction step:

Real map → graph (nodes/edges)



Euler's insight:

A trail exists if there are 0 (Eulerian circuit) or 2 vertices of odd degree

Knowledge Graph

How to build a KG from text using a LLM

Define schema

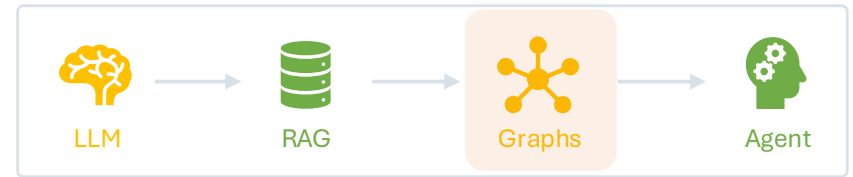
What types of information the document contains and should produce as graph elements (e.g., entities, actions, constraints, dates)

Extract relations

Use the LLM to identify entities and relationships from text according to the schema.

Insert into graph

Map extracted entities and relations into nodes and edges, then store them in the graph database.



Text → structured facts

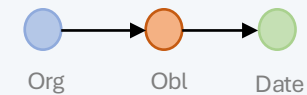
Clause text

“Organizer shall confirm headcount no later than 5 days before event.”

Extraction (JSON)

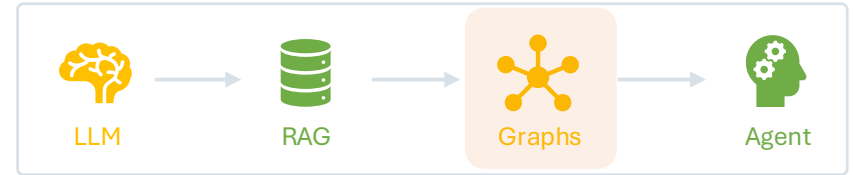
```
{
  "party": "Organizer",
  "obligation": "confirm headcount",
  "date": "19 May 2026"
}
```

KG



Graph-RAG

Base KG + retrieval enrichment for robust answers



Baseline graph

Build a core KG from the contract (parties, obligations, deadlines)

Baseline KG

From contract and annex



Enrich with retrieval

Retrieve evidence relevant to the user query and add new nodes/edges to the KG

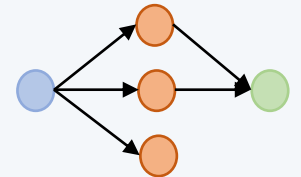
RAG retrieval

Top chunks for user query:

- Cancellation refund
- Change fees
- Venue constraints

Expanded graph

Merge with baseline KG



Query the expanded KG

Travel through the relations to compute the answer

Answering a query

Use the graph facts and retrieved chunks
Answer with citations + graph path trace

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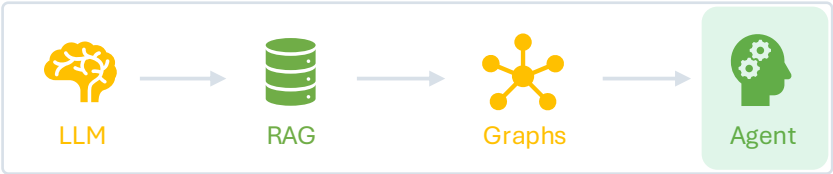
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BREAK

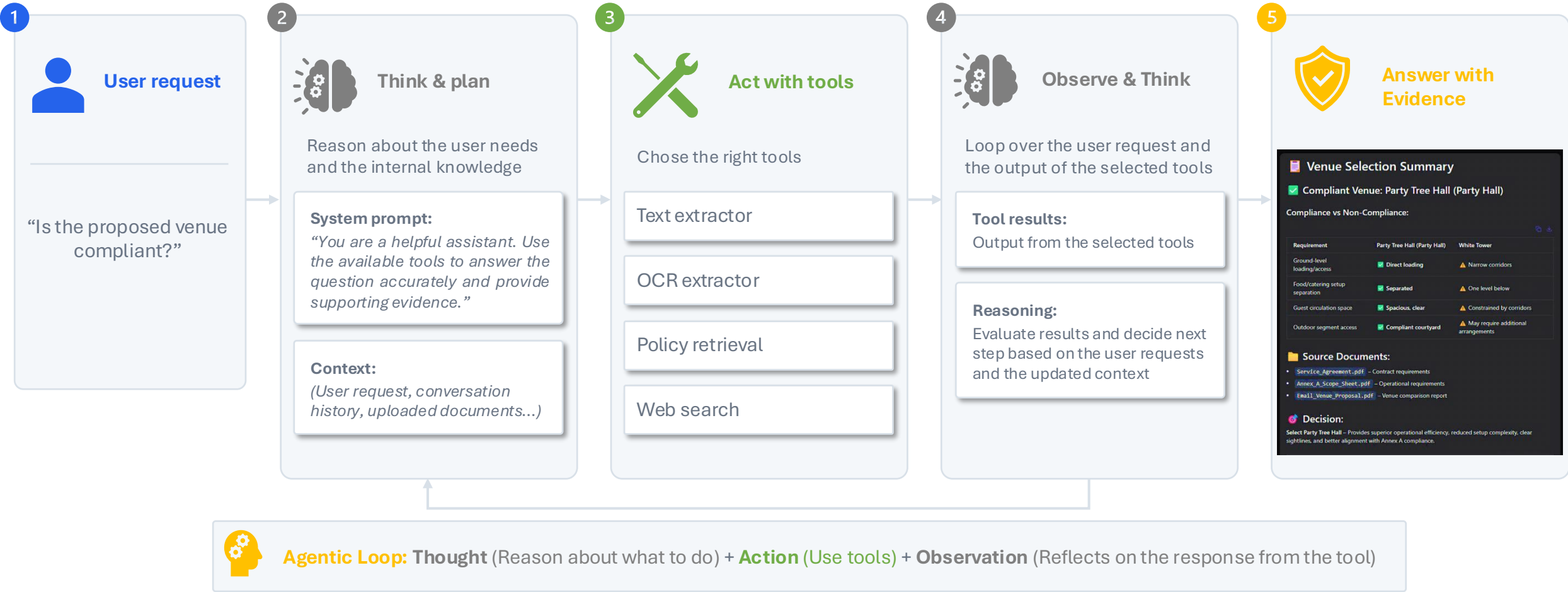
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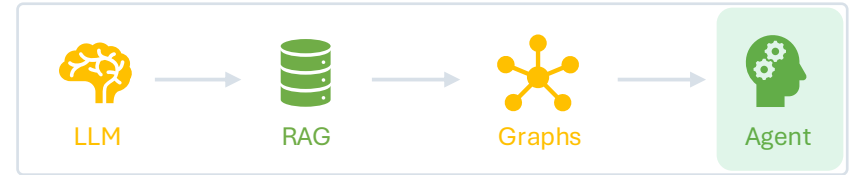


Agent

LLM-based system that can reason, choose tools, and act to complete a task.



Tools



Tools give the agent capabilities beyond the model

Definition

- A tool is a function or service that the agent can call to get information or perform an action.
- The agent decides when to call which tool based on the user's request and available tools.

Good tools

- Have clear, specific name (e.g., `get_current_time`)
- Have clear description (When to use, what it does, what it returns)
- Have a well-defined input & output schemes (Parameters the model can fill)
- Return structured, useful results (Help the agent to reason)

Think & plan

Based on the user request and available tools the model decides to call a tool

Tool calling

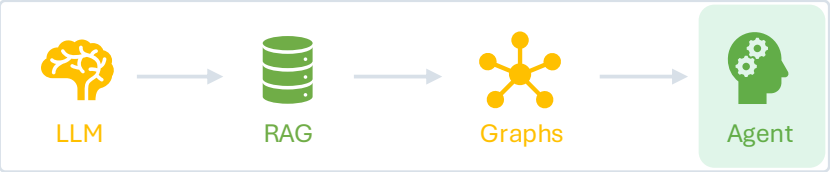
The model call the selected tools

Tool execution

The system runs the tool and returns the result

Answer

The results are added to the conversation. The model uses them to produce an answer



Alpine Event Contract Assistant

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1 Alpine Event Agreement

Service Agreement

SERVICE AGREEMENT - ALPINE EVENING

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1. Purpose
Provider shall deliver the "Alpine Evening" services (food and performance) for Organizer, as described in Annex A (Scope Sheet).

2 Annex A

Annex A - Scope Sheet

ANNEX A - SCOPE SHEET

Document ID: ELC-SHC-2026-02-A
Linked Agreement: ELC-SHC-2026-02

Event
Event date: 24 May 2026
Venue: Venue to be selected based on operational and spatial suitability for the event format, including guest circulation, catering setup, and optional live performance requirements.
Event time: 18:00-20:00
Setup access: from 16:30

Services

- Food station: 2 hours service (assorted cheeses and bread; allergen labels)
- Music: two acoustic sets (~8 minutes each)
- Optional outdoor horn segment: 10 minutes, only where the venue provides compliant and directly accessible outdoor performance space.

3 Email exchange

From: Frodo Baggins (Shire Hospitality Cooperative GmbH)

To: Arwen Undómiel (Elven Ledger Council of Rivendell SA)

Date: 06 May 2026

Subject: Potential Venue Options for Alpine Evening

Dear Arwen,

Thank you for your thorough exploration of venue options. I have reviewed both locations and would like to present my operational assessment to help inform the final selection.

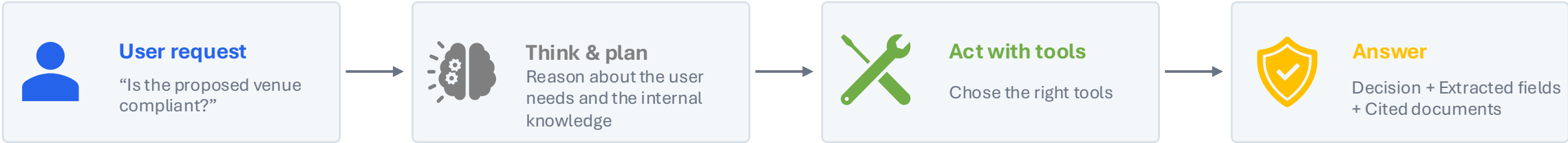
Option 1: The White Tower Upper Gallery (Minas Tirith)
This venue offers undeniable visual distinction with panoramic city views and elevated terraces providing a premium backdrop for the event. The eastern balcony location is ideal for

4 Fondue Recipe

Fribourg Fondue Moitié-Moitié
for 50 people

Ingredients:

- 5 kg Vacherin Fribourgeois AOP, grated
- 5 kg Gruyère Switzerland AOP, grated
- 5 l dry white wine
- 250 g potato starch
- 10 to 12 cloves garlic
- 20 cl kirsch (optional)
- white pepper, to taste
- freshly grated nutmeg



Goal: Combine extraction, retrieval, and useful tools to support contractual decisions

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Thank you !

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